AgroEarthFM

Complete Finalized Specification

Scope-Lock, Data-Readiness, and Comprehensive Research Blueprint

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Table of Contents

*Note: Right-click the Table of Contents and select **Update Field** to refresh page numbers after opening.*

Part I: Research Foundation

1. Research Vision and Background

1.1 The Scientific Gap

Current geospatial foundation models (GeoFMs) such as SatMAE, DOFA, PhilEO, and TerraFM learn powerful visual representations from Earth observation data, but they are fundamentally limited in the agro-ecosystem domain for three critical reasons. First, they learn appearance-based representations optimized for image reconstruction or contrastive objectives, not process-centered representations that understand how drought stress propagates through soil moisture decline, vegetation stress, phenology disruption, and ultimately crop yield reduction. Second, they treat each pixel or patch independently, ignoring the temporal causal chain that connects climate forcing to agricultural outcomes. Third, they lack physics-awareness — they can detect that a field looks stressed but cannot enforce or verify that their representations are consistent with basic water-balance or plant-growth constraints.

The central innovation of AgroEarthFM is to move beyond appearance-based geospatial representation learning toward process-centered representation learning that jointly models drought dynamics and crop phenology within physical and causal constraints. This means the model does not merely learn to reconstruct satellite imagery — it learns to predict how drought stress evolves over time, how phenology transitions respond to environmental forcing, and how these processes interact under the governing constraints of water balance and plant physiology.

1.2 Research Direction

AgroEarthFM merges two research directions that have traditionally been treated separately. Direction A involves drought modeling, where remote sensing and climate data are used to detect, monitor, and forecast drought conditions, typically through indices such as SPI, SPEI, VHI, and soil moisture anomalies. These approaches focus on the climate-hydrology-vegetation chain but rarely model crop-specific phenology or yield implications. Direction B involves crop phenology modeling, where time-series of vegetation indices are used to detect phenological transitions (green-up, heading, maturity, senescence) and predict crop development. These approaches focus on crop-specific temporal patterns but rarely incorporate drought dynamics as a process driver. AgroEarthFM unifies these directions by learning a shared multimodal temporal representation that captures both drought stress evolution and crop phenology dynamics as interacting processes, constrained by physical laws and structured by causal relationships.

1.3 Long-Term Vision (Phase 2 and Beyond)

The long-term vision (24-month horizon) extends AgroEarthFM to include: full neural causal discovery with validated counterfactual reasoning, food-security risk assessment requiring socioeconomic vulnerability and exposure data, global multi-biome generalization across diverse agro-climatic zones, integration with process-based crop simulation models (DSSAT/APSIM/AquaCrop) for hybrid physics-data modeling, and operational decision-support deployment for agricultural resilience planning. These are explicitly deferred from the first-year MVRC and will be pursued only after the core scientific contribution is validated.

2. Revised Core Hypothesis

2.1 Final Revised Hypothesis

A multimodal temporal representation trained on joint drought and crop-phenology process objectives, and regularized by weak water-balance constraints, will improve crop-stress detection, phenology forecasting, and yield-risk prediction under out-of-region, out-of-year, and extreme-event validation compared with single-domain, non-physics, and conventional machine-learning baselines.

2.2 Hypothesis Rationale

This hypothesis is structured as a conditional causal claim: IF the model is trained with joint process objectives AND regularized by weak physics constraints, THEN it will outperform baselines on specific evaluation settings. The "weak" qualifier on physics constraints is essential — the water-balance constraint is incomplete (missing irrigation, groundwater, management), and the plant-growth constraint is approximate (Mitscherlich-Baule is a simplified growth response). The hypothesis claims that even incomplete physics, applied as soft regularization with uncertainty weighting, provides measurable benefit over unconstrained models.

2.3 Hypothesis Scope Boundary

The hypothesis does NOT claim: full causal discovery from observational data alone; food-security risk prediction (requires socioeconomic data not included); global generalization across all climate zones; decision-grade counterfactual estimates; or field-level yield accuracy from administrative-level yield labels. These are explicitly excluded from the first-year MVRC scope.

3. Atomic Testable Claims

3.1 Claim Register

Claim ID	Claim	Validation Requirement	Success Signal	Failure Action
C1	Joint drought + phenology modeling improves representation quality	Compare joint vs drought-only and phenology-only models	Statistically meaningful improvement ($p < 0.05$)	Reduce model complexity; inspect task conflicts
C2	Multimodal fusion improves performance and robustness	Compare satellite-only, weather-only, soil-only, and fused models	Fused model improves OOD/extreme by >=5% relative	Check alignment, missing data, attention collapse
C3	Process-centered SSL improves transfer over generic SSL	Compare process SSL vs generic MAE/contrastive SSL	Improved transfer, fine-tuning efficiency, or extreme-event recall	Revert to generic SSL; report negative finding
C4	Weak physics regularization improves consistency without harming skill	Compare with and without water-balance/growth constraints	>=15% reduction in water-balance RMSE, <=3% task drop	Reduce physics weight; report as diagnostic
C5	Temporal causally informed structure improves interpretability	Temporal-DAG sign checks, invariance tests, known-event sensitivity	Correct signs, stable invariance, plausible scenarios	Treat as interpretability only, not performance claim
C6	The model performs better on difficult conditions	Out-of-region, out-of-year, drought-event holdouts	>=10% relative improvement in event-level F1	Rebalance training, add event-focused objectives

3.2 Claim Validation Priority

Priority	Claims	Justification
Must-validate	C1, C2, C3	Core novelty of the paper
Strongly-validate	C4, C6	Key differentiators
Validate-if-possible	C5	High-risk, high-reward; acceptable as interpretability contribution

4. Causal Chain and Temporal DAG

4.1 The Agro-Ecosystem Process Chain

The fundamental process chain that AgroEarthFM models is: Climate Forcing → Temperature Anomalies → ET Imbalance → Soil Moisture Decline → Vegetation Stress → Phenology Disruption → Crop Stress Accumulation → Yield Reduction. This chain represents the physical pathway through which drought impacts propagate through the agro-ecosystem. Each link in this chain is grounded in established agricultural and hydrological science.

4.2 Original Causal Graph (DEFECTIVE — Contains DAG Cycle)

The original implementation plan proposed a static causal graph with a bidirectional edge between Vegetation Health (VH) and Crop Stage (CS): VH ↔ CS (bidirectional, same time step). This violates the Directed Acyclic Graph (DAG) requirement because it creates a simultaneous cycle. Blocker Status: RESOLVED — replaced with temporal DAG.

4.3 Revised Temporal DAG (FINAL)

The revised causal structure is time-indexed, eliminating same-time-step cycles:

```
Weather(t) → ET(t)

Rainfall(t), ET(t), Soil(t), I_latent(t) → Soil_Moisture(t)

Soil_Moisture(t), Temperature(t), Crop_Stage(t) →
Vegetation_Health(t)

Vegetation_Health(t), AGDD(t), Crop_Calendar → Crop_Stage(t+1)

Vegetation_Health(t), Crop_Stage(t), Stress_Accumulation(t) →
Yield_Risk(t+1)
```

4.4 DAG Structural Properties

Property	Specification
Acyclicity	No same-time directed cycles; temporal indexing ensures DAG validity
Edge directionality	All edges are temporally forward or same-time with clear causal precedence
VH → CS	Forward in time: VH(t) → CS(t+1)
CS → VH	Same time step: CS(t) → VH(t) (crop stage determines stress sensitivity)
I_latent role	Latent unobserved water input enters soil moisture equation at time t
Stress accumulation	Cumulative term that accumulates across time steps

4.5 Key Causal Relationships Explained

Weather → ET: Temperature and radiation drive evapotranspiration through the Penman-Monteith relationship. Higher temperatures increase atmospheric demand for water vapor, accelerating ET rates.

Rainfall + ET + I_latent → Soil Moisture: The water balance determines soil moisture state. Rainfall and latent water inputs add water; ET and runoff remove it. The I_latent term captures all unobserved water inputs including irrigation, groundwater, and model errors.

Soil Moisture + Temperature + Crop Stage → Vegetation Health: Vegetation health depends on water availability, atmospheric stress, and the crop's development stage which determines its stress sensitivity. Crops are most sensitive to drought during flowering and grain-filling stages.

Vegetation Health + AGDD + Crop Calendar → Crop Stage (t+1): Phenology advances based on accumulated thermal time and the crop's current health status. Stressed crops may experience delayed phenology. The crop calendar provides a prior for expected transition timing.

Vegetation Health + Crop Stage + Stress Accumulation → Yield Risk (t+1): Yield risk emerges from the cumulative stress experienced by the crop, weighted by the phenology stage at which stress occurred. Stress during reproductive stages has disproportionate yield impact.

Part II: Architecture Specification

5. Complete Model Architecture

5.1 MVP Architecture (Final)

The first implementation uses a staged architecture: Fine Satellite Encoder (ViT-based) → Weather/Climate Temporal Encoder → Soil/Static Context Encoder → Crop Calendar/Crop Type Embedding → Scale-Aware Fusion Module → Hierarchical Temporal Memory Module (5-day + Monthly) → Shared Agro-Ecosystem State Representation → Three Output Heads (Crop Stress Detection, Phenology Forecast, Yield Risk/Yield Anomaly) → Weak Physics Regularization (Water Balance + Plant Growth) during training.

5.2 Component Specifications

5.2.1 Fine Satellite Encoder

Parameter	Specification
Base architecture	Vision Transformer (ViT)
Input	Sentinel-2 / HLS multi-spectral imagery (10-30m)
Patch size	16×16 pixels
Input bands	B2,B3,B4,B8 (10m) + B5,B6,B7,B8A,B11,B12 (20m resampled)
Temporal input	Sequence of 5-day composites
Positional encoding	2D spatial + temporal encoding
Approximate parameters	30-50M parameters

5.2.2 Weather / Climate Temporal Encoder

Parameter	Specification

Parameter	Specification
Base architecture	Temporal Transformer
Input	ERA5 / CHIRPS climate variables at native resolution (~31km / ~5km)
Variables	Temperature (min/max/mean), precipitation, wind speed, solar radiation, humidity
Temporal resolution	Daily aggregated to 5-day windows
Approximate parameters	10-15M parameters

5.2.3 Soil / Static Context Encoder

Parameter	Specification
Base architecture	MLP or single GNN if topology available
Input	SoilGrids variables at ~250m resolution
Variables	Clay, sand, organic carbon, pH, CEC, bulk density, soil depth
Approximate parameters	2-5M parameters

5.2.4 Crop Calendar / Crop Type Embedding

Parameter	Specification
Architecture	Learned embedding layer
Input	Crop type, planting date, expected harvest date
Source	GEOGLAM crop calendars, crop masks
Approximate parameters	<1M parameters

5.2.5 Scale-Aware Fusion Module

Parameter	Specification
Architecture	Cross-attention with resolution-aware positional encoding
Input	Token sequences from all encoders
Mechanism	Fine-resolution tokens attend to coarse-resolution context
Resolution handling	Multi-resolution tokens preserved; no forced upsampling to 10m
Missing modality	Modality-present mask + learned missing-modality token
Approximate parameters	15-25M parameters

5.2.6 Hierarchical Temporal Memory Module

Parameter	Specification
Architecture	Two-level temporal transformer
Level 1	Short-term memory: 5-day composite sequence (within-season dynamics)
Level 2	Long-term memory: Monthly aggregated sequence (seasonal patterns)
Approximate parameters	20-30M parameters

5.2.7 Output Heads

Head	Architecture	Output
Crop Stress Detection	MLP with sigmoid output	Binary/multi-class stress probability per pixel per time step

Head	Architecture	Output
Phenology Forecast	MLP with softmax over stages	Stage probability + transition timing (MAE in days)
Yield Risk / Yield Anomaly	MLP with Gaussian output	Yield anomaly prediction with uncertainty

Note: The Food-Security Risk head is REMOVED from the MVP. Food-security risk requires socioeconomic vulnerability, market, storage, and population data not included in the first model. This has been reframed as agricultural production risk or yield-shortfall risk.

5.3 Components Deferred from MVP

Component	Status	Reason	Re-evaluation Condition
Full spatial topology GNN	Optional extension	Requires robust graph construction	If spatial context encoding insufficient
Full neural causal discovery	Deferred (Phase 2)	High validation burden	After causal module passes G6 gate
Counterfactual reasoning module	Deferred or limited demo	Requires validated temporal-DAG + physics	After C5 validated and physics stable
Food-security output head	Removed from MVP	Requires socioeconomic data	If socioeconomic data pipeline built in Phase 2
Resilience score head	Deferred	Requires validated stress + recovery modeling	After crop-stress detection is robust
DSSAT/APSIM integration	Optional case-study baseline	Requires detailed management data	If process-model comparison essential

5.4 Model Size and Token Budget

Component	Estimated Parameters	Token Budget
Satellite Encoder (ViT)	30-50M	~256 tokens per time step
Weather Encoder	10-15M	~1-4 tokens per time step
Soil Encoder	2-5M	~1-4 tokens per time step
Crop Calendar Embedding	<1M	~1 token per time step
Fusion Module	15-25M	Cross-attention: fine queries, all keys/values
Temporal Memory	20-30M	5-day: ~24 tokens; Monthly: ~6 tokens
Output Heads	2-5M	Per-head MLP
Total MVP	~80-130M	~280-290 tokens per time step

6. Missing Modality Protocol

6.1 Design

The model must support missing or unavailable inputs during both training and inference. For each modality: (1) Add a modality-present mask (binary indicator per sample per modality). (2) Add a learned missing-modality token that replaces absent modality tokens.

(3) Use modality dropout during training (randomly zero out entire modalities with probability p=0.1-0.2). (4) Report performance under missing-modality scenarios.

6.2 Missing-Modality Test Matrix

Test	Description	Acceptance Criterion
Full modality	All inputs available	Baseline performance
No soil moisture	Remove SMAP/ESA CCI	<5% relative drop in primary metrics
No irrigation proxy	Remove irrigation-related features	Documented performance impact
No SAR (if included)	Test optical-only fallback	<10% relative drop
Cloud-degraded optical	Simulate missing optical observations	Graceful degradation, not catastrophic failure
Weather-only	No satellite imagery	Documented baseline for zero-optical scenarios

Part III: Data Strategy

7. Multi-Resolution Datacube Design

7.1 Core Principle

The model does NOT force all data to a common 10m grid. Each variable is represented at its appropriate spatial and temporal scale. The fusion module is designed to handle multi-resolution inputs natively. The original plan's approach of resampling all data to 10m Sentinel-2 resolution creates visually detailed but physically false precision. ERA5, SMAP, GRACE, groundwater, administrative yield, and management data do not support true 10m inference.

7.2 Resolution Tiers

Tier	Spatial Scale	Temporal Scale	Data Types	Token Treatment
Fine	10-30m	5-day composites	S2/HLS reflectance, vegetation indices, SAR	Fine-resolution visual/vegetation tokens
Intermediate	100-500m	5-day to monthly	Aggregated vegetation, SAR moisture proxies	Aggregated context tokens
Coarse	1-10km	Daily to monthly	ERA5/CHIRPS weather, SMAP/ESA CCI soil moisture	Coarse temporal weather/moisture tokens
Regional	Admin/field level	Seasonal/annual	Yield statistics, management records, crop maps	Regional context tokens

7.3 Datacube Construction Protocol

(1) Coregistration: All data within each tier reprojected to common CRS. (2) Temporal alignment: All data aligned to 5-day composite windows. (3) Cloud handling: 5-day composites use median compositing with valid-pixel filtering. (4) No cross-tier resampling: Fine-tier data NOT upsampled from coarse-tier; coarse-tier NOT forced to 10m. (5) Metadata preservation: Each variable retains native resolution, uncertainty estimate, and source metadata.

8. Complete Data Source Catalog

8.1 Satellite Imagery

Source	Product	Resolution	Bands	Access	License
Sentinel-2	L2A reflectance	10-20m	B2-B12, B8A	GEE / Copernicus Hub	CC-BY-SA 4.0
HLS	HLSL30/HLSS30	30m	Visual + NIR + SWIR	LP DAAC / GEE	Public domain
Landsat 8/9	L2 reflectance	30m	Multi-spectral	USGS / GEE	Public domain

8.2 Vegetation Indices

Index	Formula	Purpose	Derivation
NDVI	(B8 - B4) / (B8 + B4)	Green vegetation density	From S2 reflectance
EVI	2.5 × (B8 - B4) / (B8 + 6×B4 - 7.5×B2 + 1)	Corrected vegetation density	From S2 reflectance
NDWI	(B8 - B11) / (B8 + B11)	Vegetation water content	From S2 reflectance
LAI	PROSAIL inversion or empirical	Leaf area index	From S2 reflectance
FAPAR	PROSAIL inversion or empirical	Absorbed PAR	From S2 reflectance
NDMI	(B8 - B11) / (B8 + B11)	Normalized difference moisture	From S2 reflectance

Note: Phenology transitions derived from vegetation-index derivatives are noisy pseudo-labels. The plan explicitly acknowledges this and estimates label noise rather than treating them as ground truth.

8.3 Climate Data

Source	Variables	Spatial Resolution	Temporal	Access	License
ERA5	T_min, T_max, T_mean, P, wind, RH, solar, ET0	~31km	Hourly→daily→5-day	CDS API / GEE	CC-BY-4.0

Source	Variables	Spatial Resolution	Temporal	Access	License
CHIRPS	Precipitation	~5km	Daily	CHIRPS API / GEE	Public domain
ERA5-Land	Enhanced land variables	~9km	Hourly	CDS API	CC-BY-4.0

8.4 Soil Data

Source	Variables	Resolution	Type	Access
SoilGrids 250m	Clay%, Sand%, Organic C, pH, CEC, Bulk density, Depth	~250m	Static	ISRIC / GEE
ISRIC	Soil taxonomy, water holding capacity	Variable	Static	ISRIC

8.5 Soil Moisture

Source	Product	Resolution	Depth	Access
SMAP	L3 Enhanced	~9km (3km enhanced)	0-5cm	NASA DAAC / GEE
ESA CCI	Active + Passive merged	~25km	0-5cm	ESA CCI / GEE

Critical note: Soil moisture products provide coarse-resolution surface (0-5cm) estimates. They represent large-area moisture conditions, not field-scale root-zone soil moisture. They should be treated as coarse context, not fine-grained truth.

8.6 Groundwater

Source	Product	Resolution	Access
GRACE/GRACE-FO	Liquid water equivalent thickness	~150-300km	NASA JPL / GEE

Optional regional context only. Resolution is far too coarse for field-scale modeling. Used only as a regional drought indicator if included at all.

8.7 Crop Calendar and Crop Mask

Source	Variables	Resolution	Access
GEOGLAM Crop Calendar	Planting/harvest dates by crop and region	Regional/country	GEOGLAM / USDA
ESA WorldCereal	Crop type maps	10m	ESA
CDL (US)	Crop type	30m	USDA NASS
ISRO Crop Maps (India)	Crop type, kharif/rabi	State/district	Government of India / ISRO

Crop mask errors: Crop-type maps contain errors and should be treated as noisy labels. Use confidence filtering and crop-mask uncertainty. Sensitivity testing with high-confidence crop pixels only is required.

8.8 Yield Labels

Source	Scale	Access	Quality
USDA NASS (US)	County-level	Public	High quality, annually updated
ICRISAT (India)	District-level	Public	Good quality, historical series
FAOSTAT	Country-level	Public	Coarse, cross-country comparison
Field-level surveys	Field-level	Varies	Rare but valuable when available

Scale mismatch rule: If yield is available only at district/county level, field-level predictions must be aggregated to the label scale before validation. Never claim field-level yield accuracy from administrative-level labels.

8.9 Irrigation and Management Proxies

Source	Data Type	Resolution	Access
GMIA (FAO)	Irrigated area fraction	~10km	FAO
IWMI Global Irrigation Map	Irrigated/rainfed classification	~500m	IWMI
ESA WorldCereal	Irrigation classification	10m	ESA
Census data	Irrigated area by district	Admin level	Government statistical offices

Critical decision: Direct irrigation data is unreliable at fine spatial scales. The model uses I_latent (latent residual water-input inference term) rather than attempting to directly detect irrigation. Irrigation maps are used only as validation references for I_latent, not as direct inputs.

9. Scope-Lock Specifications

9.1 Crop Selection

Crop	Justification
Wheat	Globally important cereal; well-studied phenology; extensive data; drought-sensitive during reproductive stages; major crop in IGP and Mediterranean regions
Maize	Highest global production cereal; strong phenology signal from satellite; drought-sensitive during silking/grain-fill; major crop in US Corn Belt; contrasting growing season to wheat

Rationale for two crops: Two crops provide sufficient diversity for testing cross-crop transfer (Claim C2) while remaining tractable. Wheat and maize have well-characterized phenology, extensive ground-truth data, contrasting stress sensitivities, and grow in different seasons/regions.

9.2 Study Region Selection

Region	Climate	Primary Crop	Data Justification

Region	Climate	Primary Crop	Data Justification
Indo-Gangetic Plain (Punjab/Haryana, India)	Subtropical monsoon; irrigated + rainfed	Wheat (rabi)	ICRISAT district yield; ISRO crop maps; drought events 2002/2009/2014/2022; high cloud-free density in rabi; irrigation/rainfed contrast
US Corn Belt (Iowa/Illinois)	Temperate continental; predominantly rainfed	Maize	USDA NASS county yield; CDL 30m crop maps; extensive weather stations; drought 2012/2023; low cloud contamination
Iberian Peninsula (Castilla-y-León/Andalucía, Spain)	Mediterranean; water-limited; drought-prone	Wheat	Eurostat/CAP yield; ESA WorldCereal; severe drought 2017/2019/2022-23; irrigation/rainfed contrast; tests water-limited conditions

9.3 Region Selection Criteria Verification

Criterion	IGP	US Corn Belt	Iberian Peninsula
Crop map available	Yes (ISRO/Gov India)	Yes (CDL 30m)	Yes (ESA WorldCereal)
Sufficient cloud-free observations	Yes (rabi season)	Yes (summer)	Yes (spring)
Weather data for all years	Yes (ERA5/CHIRPS)	Yes (ERA5/NOAA)	Yes (ERA5)
Yield labels available	Yes (ICRISAT district)	Yes (USDA NASS county)	Yes (Eurostat/CAP)
Known drought events	2002, 2009, 2014, 2022	2012, 2023	2017, 2019, 2022-23
Irrigation information	Yes (GMIA/IWMI)	Yes (USDA FRIS)	Yes (GMIA/IWMI)
5+ years of data	Yes (2017-2023)	Yes (2017-2023)	Yes (2017-2023)

9.4 Temporal Coverage

Parameter	Specification
Study period	2017-2023 (7 years)
Rationale	Post-S2A/B launch; captures multiple drought events; overlaps with HLS and SMAP availability
Growing seasons covered	~7 seasons per crop per region
Temporal resolution	5-day composites within growing season; monthly outside
Total time steps per season	~24 five-day composites (120 days) for in-season modeling

9.5 Train / Validation / Test Split Protocol

Split	Method	Purpose	Data Fraction
Train	Random within-region, specific years	Model fitting	~60% of years (2017-2020)
Validation	Out-of-year within-region	Hyperparameter selection, early stopping	~20% of years (2021)

Split	Method	Purpose	Data Fraction
Test-In-Region	Out-of-year, different from validation	Final in-region performance	~20% of years (2022)
Test-Out-Region	Entire held-out region	Spatial transfer evaluation	1 region held out
Test-Extreme	Drought year holdout	Extreme-event robustness	Specific drought years
Test-Early-Season	Partial season observations only	No-leakage forecasting	Early-season subsets

Split locking rule: All splits must be defined BEFORE model selection and training begins to prevent data leakage. Region/year/event splits are fixed and committed to version control.

No-leakage protocol: Future satellite observations beyond forecast date are PROHIBITED. Full-season composites in early-season predictions are PROHIBITED. Yield labels from the target prediction period are PROHIBITED. Gap-filled products using future observations are PROHIBITED unless explicitly documented. Region-level normalization using held-out region statistics is PROHIBITED. All preprocessing must be fit only on training data.

9.6 Label Sources and Scales

Region	Crop	Label Source	Scale	Variables
IGP (India)	Wheat	ICRISAT VDSA	District-level	Yield (t/ha)
US Corn Belt	Maize	USDA NASS Quick Stats	County-level	Yield (bu/acre → t/ha)
Iberian Peninsula	Wheat	Eurostat/CAP	NUTS2/Province	Yield (t/ha)

Phenology pseudo-labels: Derived from NDVI/EVI time-series derivatives. These are noisy estimates, not ground truth. Crop stress labels: Defined as vegetation health anomalies below threshold (NDVI anomaly $< -1\sigma$ or VHI < 35). Scale compliance rule: Predictions validated ONLY at the spatial scale of ground truth labels.

10. Data Processing Pipeline

10.1 Pipeline Stages

(1) Download and Archive: Raw data download from APIs with version control. (2) Coregistration: Reproject to common CRS per region. (3) Cloud/Quality Filtering: Apply SCL for S2, CFMask for Landsat; threshold <20% cloud in 5-day composite. (4) 5-day Compositing: Median composite within windows; valid-pixel count recorded. (5) Index Computation: NDVI, EVI, NDWI, LAI, FAPAR computed. (6) Temporal Alignment: All sources aligned to common 5-day time axis. (7) Multi-Resolution Stacking: Data organized by resolution tier. (8) Quality Flag Generation: Per-pixel per-time-step quality flags. (9)

Normalization: Per-variable normalization using training-period statistics only. (10) Split Assignment: Each pixel-time-step assigned to train/val/test.

10.2 Cloud Contamination Protocol

Scenario	Strategy
<20% cloud in 5-day composite	Use optical data directly
20-80% cloud	Flag as partially cloud-contaminated; use with quality weight
>80% cloud	Fall back to SAR/HLS/monthly composite; flag as gap-filled
Persistent cloud (>3 consecutive composites)	Use monthly aggregation; report as cloud-gap event
Tropical/monsoon chronic cloud	Use SAR as primary; optical as secondary

11. Storage and Compute Estimates

11.1 Storage Estimates

Data Category	Per Region (7 years)	3 Regions Total
Sentinel-2 L2A	~2 TB	~6 TB
HLS	~500 GB	~1.5 TB
ERA5 climate	~50 GB	~150 GB
CHIRPS precipitation	~10 GB	~30 GB
SoilGrids (static)	~20 GB	~60 GB
SMAP soil moisture	~30 GB	~90 GB
Crop masks/calendars	~5 GB	~15 GB
Yield labels and auxiliary	~1 GB	~3 GB
Processed datacubes	~1 TB	~3 TB
Total	~3.6 TB	~10.8 TB

11.2 Compute Estimates

Task	Estimated GPU-Hours	Hardware
Datacube preprocessing	200-500 CPU-hours	Multi-core CPU
Baseline training (RF/XGBoost)	50-100 CPU-hours	CPU
Baseline training (LSTM/TCN)	100-300 GPU-hours	1×A100
SSL pretraining (MAE/future)	1,000-3,000 GPU-hours	4-8×A100
Full model training	3,000-8,000 GPU-hours	8×A100
Physics experiments	500-1,000 GPU-hours	4-8×A100
Ablation experiments	1,000-2,000 GPU-hours	4-8×A100
Transfer/robustness eval	500-1,000 GPU-hours	4-8×A100
Total estimated	6,000-15,000 GPU-hours	8×A100 (80GB)

11.3 Reproducibility Requirements

Requirement	Specification
Random seeds	Fixed seeds for all experiments; logged in config files
Software versions	PyTorch, TorchGeo, xarray, rasterio versions frozen
Data download scripts	Version-controlled scripts with product IDs and dates
Train/val/test split files	Committed BEFORE any model training
Model config files	YAML/JSON for every experiment
Experiment tracking	Weights & Biases or MLflow for all runs
Preprocessing pipeline	Fit on training data only; saved transformers applied to val/test

Part IV: Self-Supervised Learning

12. SSL Objective Specifications

12.1 SSL Objective Sequence

Stage	Objective ID	Objective	Purpose	Activation
SSL-1	L_ssl_1	Masked Spatiotemporal Reconstruction	Learn basic EO representations	Phase 1 (Warm-up)
SSL-2	L_ssl_2	Cross-Modal Masked Prediction	Learn weather-soil-vegetation relationships	Phase 2 (Fusion)
SSL-3	L_ssl_3	Future Vegetation Trajectory Prediction	Learn temporal ecosystem dynamics	Phase 2-3
SSL-4	L_ssl_4	Phenology Pseudo-Transition Prediction	Learn crop-stage timing	Phase 3 (Process)
SSL-5	L_ssl_5	Drought Stress Trajectory Prediction	Learn drought response dynamics	Phase 3 (Process)
Scenario-1	L_scenario	Physics-Constrained Scenario Augmentation	Scenario consistency regularization	Phase 5 (Post-physics)

12.2 Detailed Objective Descriptions

SSL-1: Masked Spatiotemporal Reconstruction

Objective: Randomly mask patches in satellite imagery (both spatially and temporally) and train the model to reconstruct the masked regions. Masking strategy: 75% spatial patches masked per time step (following MAE), 20% of time steps fully masked, combined random spatial-temporal masking with varying ratios. Loss: L1 reconstruction loss on masked patches only. Rationale: This foundational SSL objective learns spatial and temporal structure in satellite imagery. High masking ratio forces meaningful representations rather than simple interpolation.

SSL-2: Cross-Modal Masked Prediction

Objective: Mask one modality and predict it from remaining modalities. Implementation includes Weather → Satellite prediction (predict NDVI/EVI from weather + soil + crop calendar), Satellite → Weather prediction, and Soil → Vegetation prediction. Loss: L2 prediction loss on masked modality tokens. Rationale: Cross-modal prediction forces the model to learn relationships between climate forcing, soil conditions, and vegetation response, directly aligned with learning the causal process chain.

SSL-3: Future Vegetation Trajectory Prediction

Objective: Given vegetation state up to time t, predict the vegetation trajectory from t+1 to t+K (K=3-6 five-day steps ahead). Input: Satellite + weather + soil up to time t. Target: NDVI/EVI values for t+1 through t+K. Loss: Weighted L2 trajectory loss with increasing weight for near-future predictions. Rationale: Teaches temporal dynamics — the model must understand how vegetation evolves under current conditions to predict its future trajectory, essential for drought forecasting and phenology prediction.

SSL-4: Phenology Pseudo-Transition Prediction

Objective: Predict whether a phenology transition will occur within the next K time steps, and identify the transition type. Target: Phenology transition labels derived from NDVI/EVI derivative analysis (green-up onset, peak, senescence onset, etc.). Label quality: These are pseudo-labels, NOT ground truth. Loss: Cross-entropy for transition classification + regression for timing. Rationale: Directly teaches the model to recognize and predict phenology transitions, the key temporal events in the crop development cycle.

SSL-5: Drought Stress Trajectory Prediction

Objective: Given current multimodal state, predict the drought stress trajectory (VHI/NDVI anomaly evolution) over the next K time steps. Target: Drought index values for t+1 through t+K. Loss: L2 trajectory loss + classification loss for drought onset/severity. Rationale: Teaches the model to understand drought evolution dynamics, the core drought-modeling capability.

Scenario-1: Physics-Constrained Scenario Augmentation (NOT strict SSL)

Objective: Apply synthetic perturbations (e.g., reduce rainfall by 30%, increase temperature by 2°C) and enforce that the model's predicted response is physically consistent. Status: Reclassified from Counterfactual SSL because synthetic interventions encode designer assumptions, not real-world counterfactual behavior. It is a consistency regularization technique. Activation: Only after physics regularization is stable (Phase 5). Rationale: Enforces that the model's representations respond to environmental perturbations in a physically plausible manner.

13. Loss Function Specification

13.1 Total Training Loss

```
L_total = Σ_i α_i × L_SSL_i
        + λ_water(t) × L_water
        + λ_growth(t) × L_growth
        + λ_causal(t) × L_temporal_DAG
        + λ_calib × L_calibration
```

13.2 SSL Objective Weight Ranges

Weight	Initial Value	Range	Schedule
α_1 (Reconstruction)	1.0	0.5-2.0	High initially, decrease as other objectives activate
α_2 (Cross-Modal)	0.5	0.3-1.0	Start at 0, ramp to 0.5 in Phase 2
α_3 (Future Trajectory)	0.8	0.5-1.5	Start at 0, ramp to 0.8 in Phase 2-3
α_4 (Phenology)	0.5	0.3-1.0	Start at 0, ramp to 0.5 in Phase 3
α_5 (Drought)	0.5	0.3-1.0	Start at 0, ramp to 0.5 in Phase 3

13.3 Physics Loss Weight Warm-up Schedules

Weight	Warm-up Period	Start Value	Target Value	Schedule
λ_water(t)	5 epochs	0.0	0.1	Linear warm-up
λ_growth(t)	5 epochs (after λ_water)	0.0	0.05	Linear warm-up
λ_causal(t)	5 epochs (after λ_growth)	0.0	0.05	Linear warm-up
λ_calib	No warm-up	0.01	0.01	Constant

13.4 Training Phase Schedule

Phase	Duration (approx.)	Active Losses	Focus
Warm-up	Epochs 1-20	L_ssl_1 (reconstruction)	Learn basic EO representations
Fusion	Epochs 21-50	L_ssl_1 + L_ssl_2 + L_ssl_3	Learn multimodal and temporal relationships
Process	Epochs 51-80	+ L_ssl_4 + L_ssl_5	Learn phenology and drought dynamics
Physics	Epochs 81-100	+ λ_water × L_water + λ_growth × L_growth	Physics regularization
Causal-informed	Epochs 101-120	+ λ_causal × L_temporal_DAG	Causal consistency (if gates pass)

13.5 Gradient Monitoring

Monitor	Threshold	Action

Monitor	Threshold	Action
Total gradient norm	>100	Reduce learning rate
Per-objective gradient ratio	>10:1 between any two	Rebalance weights
Gradient cosine similarity	<0 (conflicting)	Reduce weight of conflicting objective
Loss scale divergence	Any loss >10× initial	Pause that objective; reduce weight

14. SSL Stability Monitoring

14.1 Monitored Quantities

Metric	Purpose	Collapse Indicator
Embedding variance	Detect dimensional collapse	Variance → 0
Embedding rank	Detect rank collapse	Rank → 1
Covariance spectrum	Detect spectral collapse	Single dominant eigenvalue
Nearest-neighbor diversity	Detect representation clustering	All samples map to same centroid
Objective-specific val losses	Detect overfitting or divergence	Val loss diverges from train
Gradient norms	Detect training instability	Exploding or vanishing gradients
Linear-probe performance	Detect representation quality	Probe accuracy does not improve

14.2 Collapse Response Protocol

Condition	Response
Embedding variance < threshold	Stop training; increase temperature; add noise
Rank collapse detected	Reduce model capacity; add spectral regularization
Single objective degrades all downstream tasks	Remove or downweight that objective
Negative transfer between SSL tasks	Train single-objective ablations; identify conflicting pair
SSL provides no benefit over supervised	Revert to supervised; report SSL as negative finding

Part V: Physics Framework

15. Water-Balance Constraint

15.1 Revised Water-Balance Equation

```
ΔSM = P + I_latent - ET - R - D + ε
```

Symbol	Meaning	Units	Source
ΔSM	Change in soil moisture	mm	SMAP/ESA CCI (observed)
P	Precipitation	mm/5-day	ERA5/CHIRPS

Symbol	Meaning	Units	Source
I_latent	Latent unobserved water input	mm/5-day	Inferred by model
ET	Evapotranspiration	mm/5-day	ERA5 ET0 × crop coefficient
R	Runoff	mm/5-day	Estimated from rainfall + soil properties
D	Drainage / deep percolation	mm/5-day	Estimated from soil moisture surplus
ε	Measurement/model residual	mm/5-day	Remaining unexplained variation

15.2 I_latent: Latent Residual Water-Input Inference

I_latent is a model-inferred variable representing all unobserved water inputs that cannot be explained by measured precipitation alone. It is NOT "irrigation detection." What I_latent may capture: irrigation, groundwater contribution, capillary rise, runoff model error, rainfall data error, soil moisture retrieval error, cloud/gap-filling artifacts, and crop-rooting-depth effects. Validation: Compare residual anomalies with known irrigation maps, canal command areas, and reported irrigation districts where available. Only after validation against multiple evidence sources should I_latent be interpreted as irrigation.

I_latent acceptance criteria: (1) Residual variable improves error analysis. (2) Residual variable does NOT degrade transfer performance. (3) Residual anomalies are spatially and temporally structured (not random noise). (4) Residual anomalies correlate with known irrigation districts where data available.

15.3 Physics Loss Design

```
L_water = w_data × ||ΔSM_pred - (P + I_latent - ET - R - D)||²
```

Where w_data is an uncertainty weight that is higher when data quality is good, lower when data quality is poor, and zero when data is entirely missing. The model predicts ΔSM_pred and I_latent jointly.

15.4 Scale Considerations for Water Balance

The water-balance equation is most valid at the scale at which its components are measured. At 10m, rainfall, ET, and soil moisture products are too coarse to close the water balance reliably. The physics loss should be computed at the coarsest-resolution tier (1-10km) or at the natural aggregation scale of the dominant forcing data. Fine-resolution predictions are evaluated for task performance but are not directly constrained by pixel-level water balance.

16. Plant-Growth Constraint

16.1 Growing Degree Days Accumulation

```
GDD_t = max(0, T_mean_t - T_base)
AGDD_t = Σ GDD_τ for τ from planting date to t
```

Parameter	Wheat	Maize
T_base	0°C	10°C
T_upper (optional)	30°C	30°C
Typical AGDD to maturity	1500-2000 °C·day	1200-1800 °C·day

16.2 Growth Constraint Formulation

```
L_growth = w_growth × (
  ||predicted_VHI - f(AGDD, water_availability)||²
  + ||predicted_phenology_stage - g(AGDD, stress_accumulation)||²
)
```

Where f(AGDD, water_availability) is a simplified Mitscherlich-Baule growth response function, and g(AGDD, stress_accumulation) enforces that phenology cannot advance without sufficient accumulated thermal time. Stress accumulation can delay phenology but cannot advance it beyond thermal time limits.

16.3 Plant Growth Validation Metrics

Metric	Purpose	Acceptance
Impossible-state rate	Count predictions violating basic physical logic	<1% of predictions
Phenology-AGDD consistency	Phenology predictions consistent with accumulated GDD	Correlation >0.9
Stress-under-deficit check	Healthy growth not predicted under severe deficit without I_latent	Flag violations; document

17. Physics Acceptance Criteria

The physics-regularized model is accepted ONLY if ALL of the following conditions are met: (1) Water-balance residual RMSE decreases by ≥15% relative to the non-physics model. (2) Primary task metrics do not degrade by more than 3% relative. (3) Out-of-region performance does not degrade with physics. (4) Residuals are interpretable and not random noise; they correlate with known irrigation districts where data available.

Fallback path: If physics fails acceptance criteria, physics will be reported as a diagnostic analysis rather than a core model contribution. The paper will document the physics experiment as a negative or inconclusive finding.

Part VI: Causal Framework

18. Temporal DAG Specification

18.1 Complete Temporal DAG Equations

```
Eq 1: Weather(t) → ET(t)
Eq 2: Rainfall(t), ET(t), Soil(t), I_latent(t) → Soil_Moisture(t)
Eq 3: Soil_Moisture(t), Temperature(t), Crop_Stage(t) →
Vegetation_Health(t)
Eq 4: Vegetation_Health(t), AGDD(t), Crop_Calendar → Crop_Stage(t+1)
Eq 5: Vegetation_Health(t), Crop_Stage(t), Stress_Accumulation(t) →
Yield_Risk(t+1)
```

18.2 Edge Strength Learning

In the first version, the causal graph structure is domain-specified (fixed topology), and only edge strengths are learnable. Full neural causal discovery is deferred to Phase 2. Key edges with domain-fixed directions and learnable strength coefficients include: Weather→ET, Rainfall→SM, ET→SM, I_latent→SM, SM→VH, Temperature→VH, Crop_Stage→VH, VH→CS(t+1), AGDD→CS(t+1), VH→YR(t+1), CS→YR(t+1), Stress_Acc→YR(t+1).

18.3 Causal Loss

```
L_temporal_DAG = λ_causal(t) × [
  Σ_edges ||predicted_effect − learned_strength × cause||
  + acyclicity_penalty + sign_consistency_penalty
]
```

19. Causal Validation Metrics

19.1 Causal Validation Tests

Test	Purpose	Acceptance	Failure Action
Graph acyclicity	Structural validity	No directed cycles	Redesign temporal DAG
Directional sign checks	Correct effect directions	>95% correct signs	Debug causal layer
Invariance across years	Stable edge strengths	CV <0.3 for key edges	Reduce causal claims
Invariance across regions	Stable causal relationships	Same sign, similar magnitude	Report region-specific behavior
Extreme event behavior	Capture known drought events	Correct stress signal	Inspect I_latent; check data
Irrigated vs rainfed contrast	I_latent higher in irrigated areas	Significant difference ($p<0.05$)	Reconsider I_latent interpretation

Test	Purpose	Acceptance	Failure Action
Scenario monotonicity	Plausible intervention responses	>90% monotonic scenarios	Document exceptions
Known-event case studies	Match documented events	Qualitative agreement	Document disagreement

19.2 Causal Terminology Guidelines

Allowed Terminology	Prohibited Terminology (in MVRC)
Causally informed structure	Causal foundation model
Causally structured scenario analysis	Fully causal decision engine
Temporal DAG-guided representation	Proven causal discovery
Scenario simulation	Counterfactual inference
Causal plausibility checks	Validated causal effect estimates

20. Counterfactual Module Status

20.1 Phased Counterfactual Plan

Stage	Status	Scope	Validation Requirement
Year 1 (MVRC)	Limited scenario-sensitivity tests only	Apply synthetic perturbations; check directional responses	Sign checks, monotonicity
Year 2 (Phase 2)	Formal counterfactual module	If temporal-DAG and physics pass validation	Intervention-like evidence; natural experiment validation

All Year 1 counterfactual outputs must be labeled as "scenario simulations" unless validated using intervention-like evidence. The term "counterfactual inference" is reserved for Phase 2 and requires: validated temporal DAG (G6 passed), stable physics constraints (G5 passed), at least one natural experiment or quasi-experimental validation, and sign, magnitude, and monotonicity checks passed.

Part VII: Evaluation Framework

21. Baseline Specifications

21.1 Required Baselines

#	Baseline	Purpose	Must-include?
1	Climatology / Historical Mean	Minimum skill baseline	Yes
2	Persistence Model	Temporal forecasting baseline	Yes

#	Baseline	Purpose	Must-include?
3	GDD Phenology Model	Simple crop-physics baseline	Yes
4	SPI/SPEI/VHI Drought Index Models	Drought-monitoring baselines	Yes
5	Random Forest / XGBoost	Strong tabular engineered-feature baseline	Yes
6	LSTM / TCN	Deep temporal baseline	Yes
7	Satellite-only Model	Tests value of imagery alone	Yes
8	Weather-only Model	Tests value of climate data alone	Yes
9	Multimodal No-Physics Model	Tests physics contribution	Yes
10	Multimodal No-Process-SSL Model	Tests process SSL contribution	Yes
11	Existing GeoFM Features	Tests value over general EO representations	If feasible
12	AquaCrop/DSSAT/APSIM	Process-model comparison	If feasible

22. Evaluation Split Protocol

Split ID	Split Type	Purpose	Data Used
S1	Random within-region	Basic sanity check	Random pixel-level split within training years
S2	Out-of-year	Temporal generalization	Year 2022 held out from training
S3	Out-of-region	Spatial transfer	One region entirely held out
S4	Extreme-event holdout	Drought/heat robustness	Documented drought years held out
S5	Early-season prediction	No-leakage forecasting	Only first 40/60/80 days of season
S6	Irrigated/rainfed contrast	Confounding analysis	Split by irrigation status (if available)

23. Task Metrics and Thresholds

23.1 Crop Stress Detection

Metric	Description	Minimum Threshold	Target
AUROC	Area under ROC curve	>0.80	>0.90
AUPRC	Area under PR curve	>0.60	>0.80
Event-level F1	F1 for drought event detection	>0.50	>0.70
Spatial IoU	Intersection over Union of stress maps	>0.30	>0.50

23.2 Phenology Forecasting

Metric	Description	Minimum Threshold	Target
MAE (days)	Mean absolute error of transition timing	<10 days	<5 days
Transition F1	F1 for phenology transitions	>0.60	>0.80
Stage accuracy	Phenology stage classification accuracy	>0.70	>0.85
Calibration (ECE)	Expected Calibration Error	<0.10	<0.05

23.3 Drought Forecasting

Metric	Description	Minimum Threshold	Target
RMSE (drought index)	RMSE of predicted drought index	<0.5 (standardized)	<0.3
Onset F1	F1 for drought onset detection	>0.50	>0.70
Lead-time skill score	Skill vs climatology at various leads	Positive at 20-day lead	Positive at 40-day lead

23.4 Yield-Risk Prediction

Metric	Description	Minimum Threshold	Target
RMSE (yield)	Root mean square error	<15% of mean yield	<10%
MAE (yield)	Mean absolute error	<12% of mean yield	<8%
R^2	Coefficient of determination	>0.50	>0.70
AUC (below-threshold risk)	AUC for yield below threshold	>0.75	>0.85

23.5 Representation Quality

Metric	Description	Minimum Threshold
Linear probing accuracy	Frozen representation + linear classifier	>supervised-from-scratch baseline
Fine-tuning efficiency	Performance vs. fine-tuning samples	Better than random-init at <50% data
Transfer gap	Fine-tuning performance difference across regions	Smaller than supervised baseline

23.6 Physics Consistency

Metric	Description	Acceptance Criterion
Water-balance residual RMSE	Physical consistency measure	≥15% improvement over non-physics model
Impossible-state rate	Predictions violating basic physics	<1%
Stress-under-deficit check	Healthy growth under severe deficit without I_latent	<5% violation rate

23.7 Causal Plausibility

Metric	Description	Acceptance Criterion
Sign correctness	Correct directional effects	>95% of test cases
Monotonicity	Plausible response to interventions	>90% of scenarios
Invariance (cross-year)	Stable edge strengths	CV <0.3 for key edges
Known-event agreement	Qualitative match with documented events	Documented case studies agree

23.8 Calibration

Metric	Description	Minimum Threshold
Brier score	Probabilistic calibration	<0.2 for binary tasks
ECE	Expected Calibration Error	<0.10
Reliability curves	Visual calibration check	Close to diagonal
Prediction interval coverage	% of true values within intervals	Within 5% of nominal coverage

Part VIII: Implementation Roadmap

24. 52-Week MVRC Roadmap

24.1 Phase Breakdown

Weeks	Phase	Output	Gate
1-2	Revision and scope lock	Revised hypothesis, MVRC, study regions, tasks, metrics	G0
3-8	Data audit and datacube prototype	Multi-resolution datacube for one pilot region	G1
9-12	Domain and tool strengthening	Agronomy/drought/phenology knowledge track completed	—
13-18	Baselines	Climatology, persistence, XGBoost/RF, GDD/SPI/VHI, LSTM/TCN	G2
19-26	MVP model	Satellite/weather/soil/crop encoders and fusion module	G3
27-34	Process-centered SSL	Future trajectory, phenology, drought objectives	G4
35-40	Physics regularization	Water-balance and growth-loss ablations	G5
41-45	Transfer and robustness testing	Out-of-region/year/event evaluation	—
46-49	Causal-informed diagnostics	Temporal-DAG sign/invariance/scenario tests	G6
50-52	Final integration and report	Ablation tables, error analysis, manuscript-ready results	G7

25. Implementation Gates

Gate	Name	Go Condition	No-Go Condition
G0	Scope Lock	Final crop/region/year/task/metric/data selection is testable and data exists	No available data for any region-crop combination
G1	Data Readiness	At least 2 regions have sufficient data; cloud/missingness acceptable	No reliable labels or severe missingness with no fallback
G2	Baseline Performance	At least one simple baseline beats climatology/persistence	Labels or task definition appear invalid
G3	Multimodal Fusion Value	Fusion improves over best single-modality in at least 1 primary task	Fusion adds no value after alignment/missingness fixes
G4	Process SSL Value	Process SSL improves transfer, fine-tuning efficiency, or extreme-event performance	SSL collapses or provides no measurable benefit
G5	Physics Value	Physics improves consistency (≥15% residual reduction) without major task degradation (≤3%)	Physics worsens both residuals and task metrics
G6	Causal-Informed Value	Temporal-DAG passes sign, monotonicity, and invariance checks	Scenario behavior is unstable or contradicts domain knowledge
G7	Research Readiness	Ablations support at least 1 clear contribution; results reproducible	Results are broad but inconclusive

26. Compute and Reproducibility Plan

26.1 Compute Budget

Item	Specification
Hardware	8× NVIDIA A100 (80GB) GPU node
Minimum access period	3 months of active training
Estimated total GPU-hours	6,000-15,000 GPU-hours
Data storage	~11 TB for 3 regions
Estimated cloud cost	$15,000-40,000 (AWS p4d.24xl equivalent)

Part IX: Risk and Mitigation

27. Risk Register

#	Risk	Severity	Mitigation	Fallback
R1	Overbroad scope	High	MVRC defined for 52 weeks	Narrow scope further
R2	False 10m precision	High	Multi-resolution datacube	Aggregate to coarser scale
R3	Weak ground truth	High	Select regions by label availability	Use proxy labels with documented uncertainty
R4	Causal overclaiming	High	Temporal DAG; invariance tests; cautious terminology	Treat as interpretability only

#	Risk	Severity	Mitigation	Fallback
R5	Missing irrigation data	High	I_latent latent residual strategy	Treat irrigation as documented confounder
R6	Physics loss misspecification	Med-High	Uncertainty-weighted soft constraints	Report physics as diagnostic only
R7	Compute/storage underestimation	Med-High	Define compute budget before architecture lock	Reduce model size; fewer ablations
R8	Multimodal alignment errors	Med-High	Strict data QA and visual checks	Reduce number of modalities in MVP
R9	Too many SSL objectives	Medium	Add objectives sequentially	Train objectives separately
R10	Food-security overreach	Medium	Reframed as agricultural production risk	Remove food-security claims entirely
R11	SSL collapse	Medium	Embedding variance/rank monitoring	Restart with adjusted hyperparameters
R12	Negative transfer between SSL tasks	Medium	Single-objective ablations	Remove harmful objective
R13	Cloud contamination in tropical regions	Medium	SAR fallback; monthly composites	Exclude worst-affected regions/seasons
R14	Temporal lag ambiguity	Medium	Lag-window ablations	Select lag by validation performance
R15	Crop-mask noise	Medium	Confidence filtering; sensitivity testing	Use high-confidence pixels only
R16	Yield-label scale mismatch	Medium	Aggregate predictions to label scale	Validate only at label scale
R17	Calibration missing	Medium	ECE, Brier score, reliability curves	Add calibration post-hoc
R18	Domain knowledge gap	Medium	Structured reading track (Weeks 9-12)	Consult domain experts

28. Fallback Paths Summary

Scenario	Fallback Action
Physics harms task performance	Report physics as diagnostic; claim only task performance
Causal module fails sign/monotonicity checks	Treat as interpretability tool; remove causal claims
SSL provides no benefit over supervised	Report supervised results; document SSL as negative finding
Fusion adds no value over single modality	Report single-modality results; analyze failure mode
I_latent is uninterpretable	Remove I_latent; treat irrigation as documented confounder
Multi-resolution fusion too complex	Fall back to moderate-resolution (250m-1km) single-grid approach
Drought events insufficient in data	Extend temporal window; include earlier Landsat-era data
Compute budget exceeded	Reduce model size; fewer ablations; single-region study

29. Ethical and Communication Constraints

29.1 Allowed Language

Agricultural production-risk prototype. Crop-stress monitoring model. Yield-shortfall risk estimator. Causally informed scenario analysis. Multimodal representation learning framework.

29.2 Prohibited Language (in MVRC/Year 1)

Food-security crisis prediction. Fully causal decision engine. Guaranteed climate adaptation recommendation. Field-level yield truth (if labels are administrative). Operational decision-support system. Validated counterfactual inference.

29.3 Required Disclaimers

All maps and risk outputs must include: uncertainty estimates (prediction intervals or calibrated probabilities), caveats about spatial scale limitations, statement that the model is a research prototype not an operational system, and acknowledgment of data limitations (cloud gaps, coarse resolution, missing irrigation).

Part X: Validation History and Corrections

30. Blocker Resolutions (All Resolved)

30.1 B1: DAG Cycle — RESOLVED

Original problem: Bidirectional edge VH ↔ CS at same time step violated DAG acyclicity. Resolution: Temporal DAG with VH(t) → CS(t+1) and CS(t) → VH(t) (no same-time bidirectionality). Validation: Check graph acyclicity after temporal unrolling. Fallback: Remove causal module from MVRC; keep interpretability only.

30.2 B2: Timeline Realism — RESOLVED

Original problem: 52 weeks insufficient for full plan; realistic timeline 70-80 weeks. Resolution: Define MVRC for 52 weeks; full plan as 24-month vision. Validation: Milestone review every 8 weeks. Fallback: Convert causal/counterfactual to future work section.

30.3 B3: Irrigation Data Gap — RESOLVED

Original problem: No reliable global irrigation data at fine scales. Resolution: I_latent — latent residual water-input inference term (NOT irrigation detection). Validation: Compare residual anomalies with irrigation maps where available. Fallback: Treat irrigation as documented confounder/uncertainty source.

30.4 B4: Food-Security Overreach — RESOLVED

Original problem: Food-security risk requires socioeconomic data not included in model. Resolution: Reframed as agricultural production risk / yield-shortfall risk. Validation: No food-security claims without socioeconomic data. Fallback: Add food-security layer only in Phase 2 extension.

31. High-Priority Resolutions (All Resolved)

H1 (Forced 10m harmonization): Multi-resolution datacube with scale-aware fusion adopted. No coarse variables interpreted as true 10m signals. H2 (Loss weighting unspecified): Weighted loss with curriculum scheduling; warm-up for physics and causal losses. H3 (Process-model baseline absent): Added simple physics baselines (GDD, SPI/SPEI/VHI); DSSAT/APSIM if data allow. H4 (Missing modality protocol): Modality masks + learned missing-modality tokens + modality dropout. H5 (Tier 3 validation undefined): Full metrics defined for physics and causal plausibility.

32. Medium-Priority Resolutions (All Resolved)

Issue	Resolution
SSL collapse risk	Embedding variance/rank/covariance spectrum monitoring with restart protocol
Negative transfer	Single-objective ablations; remove harmful objectives
Cloud contamination	Cloud threshold + SAR/monthly fallback; report valid-observation statistics
Temporal lag ambiguity	Lag-window ablations; select by validation performance
Crop-mask noise	Confidence filtering; sensitivity testing with high-confidence pixels
Yield scale mismatch	Aggregate predictions to label scale for validation
Calibration missing	ECE, Brier score, reliability curves, prediction interval coverage
Compute unknown	GPU-hour and storage estimates; compute budget locked before training
Domain knowledge gap	Agronomy/drought/phenology reading track in Weeks 9-12

33. Terminology Corrections

Original Term	Corrected Term	Reason
Causal foundation model	Causally informed agro-ecosystem foundation prototype	Avoids overclaiming until causal validation is strong
Food-security risk	Agricultural production risk / yield-shortfall risk	Food security requires socioeconomic data not in model
Counterfactual SSL	Physics-constrained scenario augmentation	Synthetic interventions are not strict SSL
Global cross-region generalization	Transfer across selected contrasting agro-climatic regions	Makes claim testable within available time/data

Original Term	Corrected Term	Reason
Irrigation detection	Latent unobserved water-input / residual-water anomaly inference	Residuals are not uniquely irrigation
Counterfactual inference	Scenario simulation (Year 1)	Until validated with intervention-style evidence

34. Reference Corrections

Reference	Original	Correction
Ref [10]	Reichstein et al., 2020	Year should be 2019 (Reichstein et al., Nature, 2019)
Ref [13]	Xiong et al., arXiv:2210.00035	Author should be verified — likely Xia et al.

Appendices

Appendix A: Complete Terminology Mapping

Concept	Final Term	Context
The model	AgroEarthFM	Self-Supervised Multimodal Physics-Informed Causal Geo Foundation Model
First-year scope	MVRC	Minimum Viable Research Contribution
Architecture approach	Staged implementation	Components added sequentially with gating
Causal framework	Temporal DAG	Time-indexed directed acyclic graph
Physics approach	Weak uncertainty-weighted regularization	Not hard constraints; soft penalties
SSL approach	Process-centered	Objectives designed around agro-ecosystem processes
Data approach	Multi-resolution datacube	Variables at native scales; scale-aware fusion
Irrigation handling	I_latent	Latent residual water-input inference
Counterfactual	Scenario augmentation	Year 1 only; not validated counterfactual inference
Evaluation	3-tier + calibration	Representation, downstream, physics/causal, calibration
Output scope	Agricultural production risk	Not food-security risk

Appendix B: Complete Variable Catalog

B.1 Satellite Variables

Variable	Source	Resolution	Temporal
B2 (Blue, 490nm)	Sentinel-2 L2A	10m	5-day
B3 (Green, 560nm)	Sentinel-2 L2A	10m	5-day
B4 (Red, 665nm)	Sentinel-2 L2A	10m	5-day

Variable	Source	Resolution	Temporal
B5 (Red Edge 1, 705nm)	Sentinel-2 L2A	20m	5-day
B6 (Red Edge 2, 740nm)	Sentinel-2 L2A	20m	5-day
B7 (Red Edge 3, 783nm)	Sentinel-2 L2A	20m	5-day
B8 (NIR, 842nm)	Sentinel-2 L2A	10m	5-day
B8A (Narrow NIR, 865nm)	Sentinel-2 L2A	20m	5-day
B11 (SWIR 1, 1610nm)	Sentinel-2 L2A	20m	5-day
B12 (SWIR 2, 2190nm)	Sentinel-2 L2A	20m	5-day
HLS reflectance	HLSL30/HLSS30	30m	5-day

B.2 Climate Variables

Variable	Source	Resolution	Temporal
T_min, T_max, T_mean	ERA5	~31km	Daily → 5-day
Total precipitation	ERA5/CHIRPS	~31km/~5km	Daily → 5-day
Wind speed (u, v)	ERA5	~31km	Daily → 5-day
Relative humidity	ERA5	~31km	Daily → 5-day
Solar radiation	ERA5	~31km	Daily → 5-day
ET0 (Penman-Monteith)	Computed from ERA5	~31km	Daily → 5-day

B.3 Soil Variables

Variable	Source	Resolution	Temporal
Clay content	SoilGrids	~250m	Static
Sand content	SoilGrids	~250m	Static
Organic carbon	SoilGrids	~250m	Static
pH	SoilGrids	~250m	Static
CEC	SoilGrids	~250m	Static
Bulk density	SoilGrids	~250m	Static
Soil depth	SoilGrids	~250m	Static

Appendix C: Data License Summary

Data Source	License	Access Restriction
Sentinel-2	CC-BY-SA 4.0 / Copernicus	Free, open access
HLS	Public domain (NASA/USGS)	Free, open access
Landsat 8/9	Public domain (USGS)	Free, open access
ERA5	CC-BY-4.0 (Copernicus)	Free with registration
CHIRPS	Public domain	Free, open access
SoilGrids	CC-BY 4.0 (ISRIC)	Free, open access
SMAP	NASA Earthdata	Free with registration
ESA CCI SM	CC-BY-SA 4.0	Free, open access
GRACE	NASA JPL	Free with registration

Data Source	License	Access Restriction
USDA NASS	Public domain (US Gov)	Free, open access
ICRISAT	Varies	Generally open for research
Eurostat	CC-BY 4.0	Free with attribution
CDL	Public domain (USDA)	Free, open access
ESA WorldCereal	ESA license	Free for research

Appendix D: Domain Knowledge Checklist (Weeks 9-12)

D.1 Agronomy Fundamentals

Topic	Minimum Understanding Required
Crop phenology stages	Wheat: Tillering, Stem Extension, Heading, Flowering, Grain Fill, Maturity; Maize: Emergence, V1-VT, Silking R1, Grain Fill R2-R6, Maturity
GDD / thermal time	Accumulated growing degree days; base temperatures; role in phenology prediction
Crop stress mechanisms	Drought, heat, combined stress; stage-specific sensitivity; stress accumulation effects
Yield formation	Source-sink relationships; yield components; critical periods for yield determination

D.2 Drought Science

Topic	Minimum Understanding Required
Drought types	Meteorological, agricultural, hydrological, socioeconomic
Drought indices	SPI, SPEI, VHI, PDSI, soil moisture percentiles; calculation and interpretation
Drought propagation	Climate → Soil moisture → Vegetation → Crop → Yield; time lags and memory effects
ET and water balance	Penman-Monteith ET; FAO crop coefficients; soil water balance components

D.3 Remote Sensing for Agriculture

Topic	Minimum Understanding Required
Sentinel-2 for agriculture	Band selection, spectral indices, compositing, cloud masking
Phenology from satellite	NDVI/EVI time-series analysis; transition detection methods; noise and uncertainty
SAR for agriculture	Sentinel-1 backscatter; soil moisture estimation; cloud-penetrating capability
Data harmonization	HLS harmonization approach; cross-sensor calibration; temporal gap-filling

D.4 Foundation Models and SSL

Topic	Minimum Understanding Required
Geospatial foundation models	SatMAE, DOFA, PhilEO, TerraFM, Presto, etc.

Topic	Minimum Understanding Required
SSL methods	MAE, SimCLR, DINO, BYOL; masked reconstruction vs. contrastive learning
Multimodal SSL	Cross-modal prediction; modality alignment; missing modality handling
Transfer learning	Linear probing, fine-tuning, domain adaptation, OOD evaluation

Appendix E: Key Equations Summary

E.1 Water Balance

$$\Delta SM = P + I_{latent} - ET - R - D + \varepsilon$$

E.2 Growing Degree Days

$$GDD_t = \max(0, T_{mean\_t} - T_{base})$$

$$AGDD_t = \Sigma\ GDD_\tau \text{ for } \tau \text{ from planting date to } t$$

E.3 Total Loss

$$L_{total} = \Sigma_i\ \alpha_i \times L_{SSL\_i} + \lambda_{water}(t) \times L_{water} + \lambda_{growth}(t) \times L_{growth} + \lambda_{causal}(t) \times L_{temporal\_DAG} + \lambda_{calib} \times L_{calibration}$$

E.4 Physics Loss

$$L_{water} = w_{data} \times ||\Delta SM_{pred} - (P + I_{latent} - ET - R - D)||^2$$

E.5 Causal Loss

$$L_{temporal\_DAG} = \lambda_{causal}(t) \times [\Sigma_{edges}\ ||pred\_effect - strength \times cause|| + acyclicity\_penalty + sign\_penalty]$$

Appendix F: Final Acceptable Claim (If Results Support)

AgroEarthFM demonstrates that joint drought-phenology process-centered multimodal representation learning, augmented with weak physical regularization, improves agricultural stress and phenology/yield-risk prediction under selected transfer and extreme-event settings compared with strong statistical, temporal, and non-physics baselines.

Claims NOT Allowed in First-Year Output Unless Separately Validated:

The system is a fully causal foundation model. The system predicts food-security risk directly. The model generalizes globally across all climate regions. The counterfactual outputs are decision-grade causal estimates. Field-scale yield accuracy is proven using only administrative yield labels.